

LA Concepts

Mids Topics

1.8 - Linear Transformations

- A tuple is a list of elements, an n-tuple of real numbers is called a vector.
- A Matrix transformation is where a matrix is transformed into an image by taking an input. If input is $\mathbb{R}^n$ and output is $\mathbb{R}^m$, then transformation is $T: \mathbb{R}^n \Rightarrow \mathbb{R}^m$.
- If the transformation dimension of domain and range is the same, it's called an operator.
- A matrix transformation $T(x) = Ax$ for $m \times n$ matrix, where x is a column vector $\in \mathbb{R}^n$
- If matrix A has size $m \times n$, then domain = $\mathbb{R}^n$ and range = $\mathbb{R}^m$.
- **Properties:-**
 - $T_A(0) = 0$
 - $T_A(k.u) = kT_A(u)$; Homogenous Property
 - $T_A(u+v) = T_A(u) + T_A(v)$; Additive property
 - $T_A(u-v) = T_A(u) - T_A(v)$
 - Homogenous and Additive properties are called linearly conditions (prove)
- **Standard Matrix:** A matrix that represents a linear transformation $T(x) = Ax$, here A is standard matrix. Input x is multiplied by a standard matrix to yield a transformation.
- If $e_1, e_2, \dots, e_n$ are standard basis vectors, then standard matrix is $A = [T(e_1) \mid T(e_2) \mid \dots \mid T(e_n)]$. If the standard matrix is zero, then transformation is called zero transformation.
- A transformation which maps to itself is called an identity operator.
- The coefficients of input variables in a system is the standard matrix.
- Consider two standard matrix A, B . If $A=B$ then $T_A(x) = T_B(x)$.
- **Reflection Operator:** A transformation that flips a vector across the same line. For a transformation $T(x,y)$ (for 2D) and $T(x,y,z)$ (for 3D), then reflection across:
 - x-axis: $T(x,-y)$ and xy-plane: $T(x,y,-z)$
 - y-axis: $T(-x,y)$ and xz-plane: $T(-x,-y,z)$
 - Line $y=x$: $T(y,x)$ and yz-plane: $T(-x,y,z)$
- **Projection Operator:** A transformation that projects a vector on the same line. For a transformation $T(x,y)$ (for 2D) and $T(x,y,z)$ (for 3D), then reflection across:
 - x-axis: $T(x,0)$ and xy-plane: $T(x,y,0)$
 - y-axis: $T(0,y)$ and xz-plane: $T(0,y,z)$
 - yz-plane: $T(0,y,z)$
- **Rotation Operator:** A linear transformation that rotates vector around origin/axis, preserving lengths and angles.
- Standard matrix for counter-clockwise rotation is $A = [T(e_1) \mid T(e_2)]$ where $T(e_1) = (\cos\theta, \sin\theta)$ and $T(e_2) = (-\sin\theta, \cos\theta)$
- Standard matrix for clockwise rotation is $A = [T(e_1) \mid T(e_2)]$ where $T(e_1) = (\cos\theta, -\sin\theta)$ and $T(e_2) = (\sin\theta, \cos\theta)$.

4.2 - Subspace

- Subset W of a vector space V is called subspace of V if W is a vector space closed under addition, multiplication and has zero subspace. The reason for only these properties checked is because all other properties of vector space are inherited.
 - Addition: If $u, v \in W$, then $u+v \in W$.
 - Multiplication: If $u \in W$, then $k \cdot u \in W$.
 - Zero Vector: if $0 \in W$, then it is called zero subspace (closed under Axiom 1,2).
- Sets of Symmetric, Diagonal, Triangular matrices are subspaces of M_{nn} .
- **Lines Passing Through Origin**: Let W (subspace of V) be a set of lines passing through the origin, then addition of 2 vectors in W or scalar multiplication on a vector in W keeps us on the same line passing through the origin. I.e. if $u=t(1,1)$ and $v=t(2,2)$, then $u+v = t(3,3)$ is on the same line, similarly let $k=3$ then $ku = 3 \cdot t(1,1) = t(3,3)$ is also on the same line.
- **Planes Passing Through Origin**: If u and v are vectors in plane W then $u+v$ and ku keeps us on the same plane.
- **Subset Not a Subspace:-**
 - Let $W = \{x, y\} \in \mathbb{R}^2 \mid x > 0 \text{ and } y > 0$. Then it's not a subspace of W because for $k=-1$: we get point $(-1, -1)$ which is not following conditions.
 - A set of invertible matrices is not a subspace of M_{nn} because they don't have a zero matrix
- If you have a homogenous system $Ax=0$ and its solution. Then there are 3 cases:-
 - If the solution is unique, subspace is a point (origin).
 - If there's 1 free variable, subspace is line through the origin.
 - If there's 2 free variables, subspace is a plane through origin.
 - Similarly for k free variables, subspace is a k -dimensional shape through origin.

4.3 - Spanning Sets

- Any vector (x,y,z) can be linearly combined as $k_1x_1+k_2x_2+k_3x_3 = (x,y,z)$, where k_1,k_2,k_3 are some scalars.
- If the number of vectors is greater than the dimension of the space, the set is linearly dependent.
- If we have two vectors u,v and we have to check if a given vector w is a linear combination of them i.e $w = k_1u+k_2v$, then write vectors u,v as columns in matrix and augment with matrix w by writing as column. If solution exists, they it's linear combination, else it's not
- A set is a spanning set of a vector space if every vector in the space can be expressed as a linear combination of each other. So if I have a vector space V and spanning set S , then any vector of V can be written as a linear combination of vectors in S .
- Span Test:** For given vectors set: write them as rows in matrix and then:-
 - If it's a square matrix: find determinant. If its 0 then set doesn't span but if its not 0 then the given set spans vector space
 - Else reduce the matrix in RREF and if rank of RREF is same as dimension of vector space, then the set spans vector space.
- The standard basis set always spans the space.

Standard basis for $\mathbb{R}^3$: $\{(1, 0, 0), (0, 1, 0), (0, 0, 1)\}$

Standard basis for P_3 : $\{1, x, x^2\}$

Standard basis for $M_{2 \times 2}$: $\left\{ \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \right\}$

- Similarly, a set of polynomial spans a space P if every vector vector in space can be written as linear combination of polynomials in set i.e. $\text{span}\{p_1,p_2,p_3\} = \{k_1p_1+k_2p_2+k_3p_3\}$, for a polynomial space P^3 .
- You can also write the polynomials as rows in the matrix and do the same test.
- For a matrix it's also the same: for a matrix space M_{22} , write each matrix as a column vector in the matrix and do a span test.
- Two sets S and S' of vectors can span the same space if vectors of S' can be expressed as a linear combination of vectors in S and vice versa (i.e both are linearly dependent to each other).
- To prove the above point: write the first span set as column vectors as matrix and find linear dependency for all vectors in the second set. Then write the second span set as columns and find dependency. When linear combinations for both are found, it's verified.

4.8 - Row Column Null Space

- For a $n \times n$ matrix: its row vectors are row elements written normally and its column vectors are column elements written as transpose.
- The set of row vectors is called **row space** and the set of column vectors is called **column space**.
- In a linear system $Ax=b$, A takes input from x and maps b . Vectors which map to zero vectors are called **null space or kernel**. I.e. for $AX=0$, X is a null space.
- To find null space, reduce the matrix in RREF then express the solution as free variables, that solution set is null space, and the solution vectors are called **basis for null space**.
- We know that $Ax=0$ is homogenous and $Ax=b$ ($b \neq 0$) is non homogenous. Now if X_h is any solution of $Ax=0$ (i.e. Null space), and X_b is a particular solution to homogenous $Ax=b$. Then the general **solution** of non homogenous ($Ax=b$) = $X_h + X_b$. So now if you have one particular solution and null space, you can generate all solutions.
- The X_h is called subspace through origin and X_b is translation of that subspace
- **Basis for Row Space** are linearly independent rows of matrix which can be found by reducing matrix in RREF and then writing non-zero rows.
- **Basis for Column Space** are linearly independent columns of matrix which can be found by reducing matrix in RREF and then the columns which are pivot, their corresponding column from original matrix is written.
- The dimension of basis for both row and column spaces is the rank of the matrix, or the number of vectors in the basis set.
- **Basis:** Set of linearly independent vectors which span a space. A span set can contain both linearly dependent or independent vectors but the basis only contains independent.
- The dimension of the Span set is the number of vectors in the basis.
- Basis for a set can be found by writing those sets as columns in matrix then finding its basis for column space.
- If you are confused about the thing that in both span and basis you do RREF. Usually you are given a set and have to check if it spans space so you RREF for it. But for basis you are given a set of vectors from a space and you RREF it to find a basis which is column space.
- When you find the basis of a set, all the vectors not in basis can be written as a linear combination of vectors in basis.

4.9 - Rank and Nullity

- The dimension of both row and column space is called rank. If the determinant of a matrix is 0, then rank = rows, else rank < rows.
- The dimension of null space is called **nullity**.
- $\text{rank}(A) + \text{nullity}(A) = \text{variables}(A)$, the variables are unknowns or columns.
- We know that the set of all X that give $Ax=0$ is null space, but the set of all Y that give $\text{transpose}(A).y = 0$ is called **left null space**. The vector y is orthogonal to every row of A .
- Row Space is orthogonal to Null Space and Column space is orthogonal to Left null space. So they are called **Orthogonal Compliments**.
- For a $m \times n$ matrix, there are 4 fundamental spaces:-
 - Row Space $R(A)$ where each row is a vector in R^m .
 - Column Space $C(A)$ where each column is a vector in R^n .
 - Null Space $N(A)$ where each row is a vector in R^m .
 - Left Null Space $\text{trp}(A).y$ where each column is a vector in R^n .
 - Since we know that $\text{rank}(A) + N(A) = n$, similarly $\text{rank}(A) + L.N(A) = m$.
- Dimension of Null space = no of columns - Rank(A).
- Dimension of Left Null space = no of rows - Rank(A).
 - if rank = 0 then row/col space is origin
 - if rank = 1 then row/col space is line through origin
 - if rank = 2 then row/col space is plane through origin
 - If rank = k then row/col space is plane through k th-dimension.
- When you do RREF operations on a matrix to find its null space, you will have an all zero row(s). Now instead of transposing the original matrix and again doing RREF, you can simply replicate those RREF operations on Identity matrix and the row(s) corresponding to the all zeros row of original RREF, It's the left null space

Finals Topics

5.1 Eigenvalues and Eigenvectors

- An eigenvector of an $n \times n$ matrix is a non-zero vector x such that $Ax = \lambda x$ where λ is a scalar called eigenvalue of A .
- A matrix system with only trivial solutions can't have $\lambda=0$. Also a matrix is invertible if and only if it doesn't have $\lambda=0$.
- You can check if a vector is an eigenvector by multiplying it with matrix A and if A only scales, then it's an eigenvector and the value by which it scales is called eigenvalue.
- **Characteristic Equation:** $Ax = \lambda x \Rightarrow Ax - \lambda x = 0 \Rightarrow x(A - \lambda I) = 0$. But λ is scalar so to subtract from A we multiply with the Identity matrix " I " of A . Hence $(A - \lambda I)x = 0$.
- To find all eigenvectors of a matrix from a given eigenvalue, Solve the system of characteristic equations and the solution space are eigenvectors.
- **Eigenspace:** It's the set of all eigenvectors. If eigenspace has 1 free variable then the eigenspace is a line, if 2 free variables then plane and if k free variables then n -D.
- If the determinant of $A - \lambda I$ is 0, then λ is an eigenvalue. Else It's not an eigenvalue.
- To find all eigenvalues of a matrix, solve $|\lambda I - A| = 0$ which will yield multiple λ s.
- Eigenvalues of the upper triangular matrix and lower triangular matrix are its diagonals.
- **Basis for Eigenspace:** Find all eigenvalues of the matrix, then for each eigenvalue, find eigenspaces by RREF-ing $\lambda I - A$ and write them in a set, just like you find the basis of a set of vectors. Also it's impossible that no free variables are found.

5.2 Diagonalization

- **Similarity:** If two matrices are similar, then they have the same determinant, rank, nullity, trace, eigenvalues. Trace is the sum of diagonals of a matrix.
- **Similarity Transformation:** $A \Rightarrow P^{-1} \cdot A \cdot P$, is a transformation where a diagonalizable matrix can be mapped with invertible matrix P . This mapping preserves multiple properties.
- Hence a matrix B is similar to A if there exists an invertible matrix P i.e. $B = P^{-1} \cdot A \cdot P$.
- **Diagonalizable:** A $n \times n$ matrix is diagonalizable if there exists an invertible matrix P i.e. $P^{-1} \cdot A \cdot P$ is a diagonal matrix. Then P diagonalizes A .
- To diagonalize a matrix A :-
 - Find eigenvalues ($\lambda_1, \lambda_2, \dots, \lambda_n$) of A and form $\lambda I - A$.
 - Find eigenvectors $P_1 = (A - \lambda_1 I)v_1, P_2 = (A - \lambda_2 I)v_2, \dots, P_n = (A - \lambda_n I)v_n$, where v_i are unknowns. (you just write unknowns to form a system, then you solve the system and its solution is the eigenvector). Also when you have a 3×3 matrix and only 2 λ s, then either λ will give 2 solutions so you will have 3 eigenvectors.
 - Form matrix P by writing each P_i as a column.
 - Form diagonal matrix D where each leading diagonal are eigenvalues.
 - Hence P diagonalizes A and produces $D = P^{-1} \cdot A \cdot P$.
- $A = P \cdot D \cdot P^{-1}$ and $D = P^{-1} \cdot A \cdot P$ are the same.
- Matrix A can be raised to any power k by only raising diagonals of D to k and finding $A^k = P \cdot D^k \cdot P^{-1}$
- The degree of the characteristic equation is the size of the matrix (and vice versa).
- **Multiplicities:** The number of times a λ appears is called Algebraic Multiplicity (AM) and the number of vectors that appear multiple times appearing λ is called Geometric Multiplicity (GM). $1 \leq GM(\lambda) \leq AM(\lambda)$
- The dimension of eigenspace of a λ is the value = GM or $\leq AM$.
- If $GM(\lambda)$ is not equal to $AM(\lambda)$, then the matrix is not diagonalizable.

6.1 Inner Products

- A function that takes 2 vectors as input and returns scalar as output is called inner product. Just like dot products.
- A real vector space which has an inner product is called real inner product space.
- If $u, v, w \in V$ (vector space) and $k \in \mathbb{R}$, then following axioms are satisfied :-
 - Symmetry: $\langle u, v \rangle = \langle v, u \rangle$
 - Additivity: $\langle u+v, w \rangle = \langle u, w \rangle + \langle v, w \rangle$
 - Homogeneity: $\langle k.u, v \rangle = k\langle u, v \rangle$
 - Positivity: $\langle v, v \rangle \geq 0$ iff $v=0$.
- If V is a real inner product space, then **norm**(length) of a $v = |v| = \text{root}(\langle v, v \rangle)$ and distance ($d(u, v)$) b/w 2 vectors $u, v = |u-v| = \text{root}(\langle u-v, u-v \rangle)$.
- If $u, v \in V$ (real inner product space) and $k \in \mathbb{R}$, then :-
 - $|v| \geq 0$
 - $|kv| = k|v|$
 - $d(u, v) = d(v, u)$
 - $d(u, v) \geq 0$
- **Euclidean Inner Product**: If a real inner product space V is defined on $\mathbb{R}^n$, then for $u, v \in W$: $u = u_1, u_2, \dots, u_n$ and $v = v_1, v_2, \dots, v_n$. Then the euclidean inner product is $\langle u, v \rangle = u_1v_1 + u_2v_2 + \dots + u_nv_n$. If you assign weights $W = w_1, w_2, \dots, w_n$ to the inner product, then it forms a weighted euclidean inner product. $\langle u, v \rangle = w_1u_1v_1 + w_2u_2v_2 + \dots + w_nu_nv_n$. These both inner products satisfy all above axioms.
- **Standard Inner Product on Matrix**: Matrix $u, v \in M_{nn}$ then $\langle u, v \rangle = \text{trace}(u^t.v)$ i.e. sum of diagonals of the product of transpose of u with v . So $|u| = \text{root}(u_1^2 + u_2^2 + u_3^2 + u_4^2)$ and $d(u, v) = \text{root}(|u-v|)$. If $A = \text{Identity matrix}$ then it's called standard euclidean inner product.
- If $u, v \in M_{nn}$ and its inner product is generated by A , then $\langle u, v \rangle_A = u^t.A.v$. If A is diagonal then $\langle u, v \rangle_A = \text{trace}(u^t.v.w)$
- **Standard Inner Product on Polynomial**: If $p_1 = a_0 + a_1.x + a_2.x^2 + \dots + a_n.x^n$ & $p_2 = b_0 + b_1.x + b_2.x^2 + \dots + b_n.x^n$. Then it's $\langle p, q \rangle = a_0b_0 + a_1b_1 + \dots + a_nb_n$.
- $|p| = \text{root}(a_0^2 + a_1^2 + \dots + a_n^2)$ and $d(p, q) = \text{root}(|p-q|)$
- Evaluation Inner product is $\langle p, q \rangle = p(x_0).q(x_0) + p(x_1).q(x_1) + \dots + p(x_n).q(x_n)$. Here coefficients do not matter and points $x_0, x_1, \dots, x_n$ are given.

6.2 Angles and Orthogonality

- Angle between vectors is $\theta = \cos^{-1} (u.v/|u|.|v|)$. But for real inner product we need $-1 \leq \langle u,v \rangle / (|u|.|v|) \leq 1$ which is called **cauchy-schwartz inequality**.
- If $u,v,w \in V$ and $k \in \mathbb{R}$, then this inequality has 2 properties :-
 - $\|u+v\| \leq \|u\| + \|v\|$
 - $d(u,v) \leq d(u,w) + d(w,v)$
- If $u,v \in V$ and $\langle u,v \rangle = 0$, then u,v are orthogonal (because $\theta = \cos^{-1}(0) = \pi/2$)
- A set of two or more vectors is orthogonal if all pairs of distinct vectors in it are orthogonal, and a set of orthogonal vectors where each vector has norm=1 is called orthonormal.
- For matrices, 2 vectors are orthogonal if their standard inner product is 0 i.e. $\text{trace}(u^t.V) = 0$.

6.3 Orthogonal and Orthonormal Sets

- An orthogonal set of non-zero vectors is always linearly independent
- Normalizing:** Process of converting orthogonal set into orthonormal. Here we multiply each vector in a set with a reciprocal of its magnitude.
- In an inner product space, a basis consisting of orthonormal vectors is called orthonormal basis, and a basis consisting of orthogonal vectors is called orthogonal basis. Every non-zero, finite-dimensional inner product space has an orthonormal basis.
- Standard basis with euclidean inner product of $\mathbb{R}^n = \{(1,0,\dots,0), (0,1,\dots,0), \dots, (0,0,\dots,1)\}$ and of $P_n = \{1, x, x^2, \dots, x^n\}$
- Coordinates relative to basis:** If $S = \{v_1, v_2, \dots, v_n\}$ and you want to write a vector u as a linear combination i.e. $u = c_1v_1 + c_2v_2 + \dots + c_nv_n$ then.
 - If S is an orthogonal basis for V and $u \in V$, then $u = (\langle u, v_1 \rangle / \|v_1\|^2) \cdot v_1 + (\langle u, v_2 \rangle / \|v_2\|^2) \cdot v_2 + \dots + (\langle u, v_n \rangle / \|v_n\|^2) \cdot v_n$
 - If S is an orthogonal basis for V and $u \in V$, then $u = \langle u, v_1 \rangle \cdot v_1 + \langle u, v_2 \rangle \cdot v_2 + \dots + \langle u, v_n \rangle \cdot v_n$
 - By the way from this same method you can find orthogonal projection of u on w
- Coordinate Vector Relative to S:** If $S = \{v_1, v_2, \dots, v_n\}$
 - If S is an orthogonal basis for V and $u \in V$, then $(u)_S = (\langle u, v_1 \rangle / \|v_1\|^2), (\langle u, v_2 \rangle / \|v_2\|^2), \dots, (\langle u, v_n \rangle / \|v_n\|^2)$
 - If S is an orthogonal basis for V and $u \in V$, then $(u)_S = \langle u, v_1 \rangle, \langle u, v_2 \rangle, \dots, \langle u, v_n \rangle$
- So from the coefficients of writing a linear combination from the basis, we can construct a coordinate vector by writing them as components.
- Gram-Schmidt Process:** It converts arbitrary basis $\{u_1, u_2, \dots, u_n\}$ into an orthogonal or orthonormal basis $\{v_1, v_2, \dots, v_n\}$:-
 - $v_1 = u_1$
 - $v_2 = u_2 - (\langle u_2, v_1 \rangle / \|v_1\|^2) \cdot v_1$
 - $v_3 = u_3 - (\langle u_3, v_1 \rangle / \|v_1\|^2) \cdot v_1 - (\langle u_3, v_2 \rangle / \|v_2\|^2) \cdot v_2$

- $v_n = u_n - (\langle u_n, v_1 \rangle / \|v_1\|^2) \cdot v_1 - (\langle u_n, v_2 \rangle / \|v_2\|^2) \cdot v_2 - \dots - (\langle u_n, v_{n-1} \rangle / \|v_{n-1}\|^2) \cdot v_{n-1}$
- To find an orthonormal basis from it, normalize these vectors.
- QR-Decomposition:** If A is $m \times n$ matrix with linearly independent columns, then it can be factored as $A=QR$. Here Q is a $m \times n$ matrix with orthonormal column vectors and R is a $n \times n$ invertible upper triangular matrix
 - Extract columns from A to form individual vectors
 - Make them orthonormal by Gram-Schmidt Process
 - Write those orthonormal vectors as columns in matrix Q
 - Matrix R will be formed by

$\langle u_1, q_1 \rangle$	$\langle u_2, q_1 \rangle$	$\langle u_3, q_1 \rangle$
0	$\langle u_2, q_2 \rangle$	$\langle u_2, q_3 \rangle$
0	0	$\langle u_3, q_3 \rangle$
 - Write $A=QR$

7.2 Orthogonal Diagonalization

- If A and B are $n \times n$ matrix, then B is orthogonally similar to A if there exist orthogonal matrix P such that $B = P^{-1} \cdot A \cdot P$
- A square matrix $A \in \mathbb{R}^{n \times n}$ is called orthogonal if its columns form an orthonormal set.
- If A is an $n \times n$ matrix then it is symmetric, orthogonally diagonalizable and it has an orthonormal set of n eigenvectors.
- If A is symmetric, then eigenvalues of $A \in \mathbb{R}$ and eigenvectors from diff eigenspaces from different eigenspaces are orthogonal.
- Procedure to Orthogonally Diagonalize:**
 - Find Basis for each eigenspace of A
 - Apply Gram-Schmidt to each basis to obtain an orthonormal basis for each eigenspace.
 - Form matrix P from obtained orthonormal basis.
 - This matrix P will orthogonally diagonalize A and Eigenvalues on Diagonal of $D=P^t \cdot A \cdot P$ will be in the same order with respect to eigenvectors on P .
- Spectral Decomposition:** We know that if A is $n \times n$ then it's orthogonally diagonalized by $P = [u_1 \mid u_2 \mid \dots \mid u_n]$ which are orthonormal bases of A , and also $D = P^t \cdot A \cdot P$. Then spectral decomposition is $D = \lambda_1 \cdot u_1 \cdot u_1^t + \lambda_2 \cdot u_2 \cdot u_2^t + \dots + \lambda_n \cdot u_n \cdot u_n^t$
- If we are given input $x = [x_1 \ x_2 \ \dots \ x_n]$, then we can map an image by doing $D = \lambda_1 \cdot u_1 \cdot u_1^t \cdot x + \lambda_2 \cdot u_2 \cdot u_2^t \cdot x + \dots + \lambda_n \cdot u_n \cdot u_n^t \cdot x$

7.3 Quadratic Forms

- If A is a $n \times n$ symmetric matrix and X is a $n \times 1$ column vector of variables $x_1, x_2, \dots, x_n$, then a function $Q_A(x) = x^t A x$ is called a quadratic form associated with A .
- Consider a diagonal matrix A with leading entries $\lambda_1, \lambda_2, \dots, \lambda_n$. Then $x^t A x = \lambda_1 x_1^2 + \lambda_2 x_2^2 + \dots + \lambda_n x_n^2$.
- A quadratic form is a spectral decomposition where the orthonormal basis vector u is $n \times 1$.
- When you are given an equation, to find its quadratic form :-
 - Form vector X consisting of its variables.
 - To form matrix A , the coefficient of squared/cubic terms will be diagonals.
 - Off-diagonal terms are the coefficient of cross-product terms halved.
- **Principal Axes Theorem:** If A is a $n \times n$ matrix then there's an orthogonal change of variable that transforms the quadratic form $x^t A x$ into $y^t D y$ with no cross products.
- If P orthogonally diagonalizes A , then making a change of variable $x = Py$ in quadratic form $x^t A x$ yields $x^t A x = y^t D y = \lambda_1 y_1^2 + \lambda_2 y_2^2 + \dots + \lambda_n y_n^2$, where λ s are eigenvalues of A corresponding to eigenvectors with respect to P .
- To find orthogonal change of variable from given equation :-
 - Do orthogonal diagonalization of A to find matrix P and D .
 - Define $x = PY$
 - Substitute $x^t A x \Rightarrow (PY)^t A (PY) = Q$ which will become $y^t D y$.
 - Now form $Q = \lambda_1 y_1^2 + \lambda_2 y_2^2 + \dots + \lambda_n y_n^2$.

9.4 $A = CR$ and SVD Decompositions

- If A is $m \times n$ matrix and $\lambda_1, \lambda_2, \dots, \lambda_n$ are eigenvalues of $A^t A$, then $\sigma_1 = \text{root}(\lambda_1)$, $\sigma_2 = \text{root}(\lambda_2)$, $\dots$, $\sigma_n = \text{root}(\lambda_n)$ are called **singular values** of A .
- $A^t A$ is orthogonally diagonalizable and its eigenvalues are non-negative real numbers.
- Both A and $A^t A$ have the same fundamental spaces and same rank.
- In non square matrices, main diagonals can be found by starting from the top-left corner and going down diagonally till the matrix ends. Obtained values are leading diagonal.
- **Singular Values Decomposition:** $m \times n$ matrix A can be decomposed as $A = U \Sigma V^t$, where U is $m \times m$ orthogonal matrix whose columns are left singular vectors and V is $n \times n$ orthogonal matrix whose columns are right singular vectors. Σ is a diagonal matrix whose entries are singular values of $A^t A$. To find it :-
 - Find singular values of $A^t A$
 - Find right singular vectors $V_1 = (A^t A - \lambda_1 I).V_1$, $V_2 = (A^t A - \lambda_2 I).V_2$, $\dots$, $V_n = (A^t A - \lambda_n I).V_n$ and normalize these vectors.
 - Find left singular vectors $U_1 = A.V_1/\sigma_1$, $U_2 = A.V_2/\sigma_2, \dots$, $U_n = A.V_n/\sigma_n$,

- Form matrix $V = [V_1 \mid V_2 \mid \dots \mid V_n]$ and $U = [U_1 \mid U_2 \mid \dots \mid U_n]$.
 - Form matrix Σ by putting singular values in main diagonal.
 - Write $U\Sigma V^t$.
- When we have 3×3 U but only 2 eigenvalues because matrix A is 3×2 , then the third left singular vector can be found by computing the cross product of U_1, U_2 and dividing it by the magnitude of the cross product.
- **A=CR Decomposition:** Compute RREF of $m \times n$ matrix A . R = RREF form of A without zero rows and C is the basis for column space. C is $m \times r$ and R is $r \times n$, where r is rank of A .